

Mamadou THIAO, PhD

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Senior C++ Software Engineer | Applied Mathematics | Algorithms, Optimization & Performance

Summary: Software engineer & architect with extensive experience developing, building and reviewing complex software, algorithms, mathematical models, and data-processing systems across medical imaging, industrial tools, autonomous vehicles, dental health, geoscience, and scientific computing • Work through advanced reasoning, careful technical judgment, precise problem solving, and clear communication with experts • Hands-on experience with performance optimization, robust memory management, advanced concurrency and parallelization in C++ design and development • Design and develop sparse modeling solutions based on our own approach, covering feature selection in machine learning and data mining, sparse recovery in image and signal processing, portfolio selection in finance, and surface reconstruction in computer vision.

Selected open-source projects

l0l2 solver: a header-only C++ Library for Sparse Modeling using Biconjugate Convex Relaxation.

- addresses solutions for sparsity regularized linear least-square problems.
- implements our own method and several other methods.
- exploits conjugate & biconjugate functions, lagrangian duality, lower bounds, optimality conditions, Gauss-Jordan.
- offers several solvers: linear, coordinate descent, and branch-and-bound.

sparsepc solver: a header-only C++ Library for Sparse PCA.

- implements our own method and several other Sparse Principal Component and deflation methods.

Sparsely application: an standalone application software for Sparse Principal Component computation and selection**.**

- offers a very simple interface to compute sparse principal components via various methods.
- allows to run several principal component rounds.
- offers the possibility of using classical principal components as controls on the explained variances.
- aims to be very practical for engineers, scientists and other data professionals.
- eases the use of the sparse principal component analysis.

Selected Publications, Talks & Workshops: Efficient approaches for ℓ2-ℓ0 regularization and SVM at Applied Intelligence (Springer) • Sparse PCA at ICML • ℓ0 norm programs at MCO • Sparse Approximation at AFG • Sparse FDA / LDA at EURO • Feature Selection at EURO • Sparse Logistic Regression at ROADEF • Workshops on Optimization and Machine Learning • Technical Day on Autonomous Vehicles and Mobile Ground Robotics • Technical Day on Polyhedra and Combinatorial Optimization

Languages, tooling, quality & standards: Modern C++ • Python • Qt • ZeroMQ • CMake • Conan • Git • GitLab • GitHub, Jenkins • TeamCity • Visual Studio • Eclipse • Cppcheck • Valgrind

Performance engineering: Parallelization • Concurrency • Memory Optimization • Benchmarking • Multithreading • OpenMP • CUDA • Sparse Matrices

Statistics, Optimization & ML: Classification • Regression • Regularization • Maximum Likelihood • Least Squares • Robust Estimation • Convex and DC Optimization • PCA and Sparse PCA • Feature Selection • FDA/LDA • Logistic Regression • SVM • KMeans • t-SNE • SMO • Scikit-learn • Eigen • Ceres • Cplex Optimizer • Xpress Optimizer • SciPy • NumPy

Computational Geometry & CV: 3D • Point clouds • Meshes • Maps • Transforms • Surface Reconstruction • Photogrammetry • Nurbs • OpenNurbs • Eigen • VTK • Meshlab • FLANN

Software quality: unit and integration testing • continuous integration • code review • pair programming • technical documentation • cross-functional expert collaboration • engineering mentorship

Experience

Indilo – Software Engineer & Architect | Thiès, Sénégal | may 2025 – july 2026

- Prototyped scientific software from initial technical concepts through architecture, design, and implementation.
- Technologies: C++, Qt, Eigen, Python, PyQt, PyQtGraph, Scikit-learn, Scipy, Numpy, Matplotlib.

Solent – Software Engineering & Architecture Consultant | Nantes, France | june 2022 – march 2025

- **at Intrasense – Medical Imaging | june 2023 – march 2025**
- Maintained complex medical imaging software involving APIs, SDKs, DICOM, image data, image processing, computational geometry, and 3D visualization.
- Fixed defects and extended volume registration, MPR plane manipulation, and visualization capabilities.
- Improved registration processing time and resolved issues involving DICOM database files, medical imaging modalities, 3D visualizations, and application tabs.
- Delivered new add-ons, managed add-on interfaces, communicated directly with clients, and provided technical support.
- Technologies: C++, MFC, PACS, DICOM, file-based database, Visual Studio, Git, Azure, Jira.

- **at Desoutter Tools – Industrial tools | june 2022 – march 2023**
- Developed software enhancements for intelligent industrial tightening tools.
- Maintained and re-architected tightening modules and developed new tightening features.
- Tested software directly on tools and prototypes.
- Served as Scrum Master on a rotating basis.
- Technologies: C++, UML, Rhapsody, Eclipse, Arm DS-5.

Navya Group – Self-Driving Cars – R&D Engineer | La Défense, France | january 2019 – october 2021

- Developed, refactored, and integrated a NURBS processing modules, and mentored engineers on implementing NURBS processing based on papers and books.
- Refactored, optimized, and benchmarked point-cloud normal-processing modules and implementations based on papers.
- Investigated state-of-the-art approaches for storing, loading, and saving 3D maps and point clouds.
- Contributed actively to the architectural redesign of software pipelines and proposed technical solutions.
- Resolved long-standing defects and architectural issues.
- Advanced to Software Architect for the perception division, supporting mapping, localization, and obstacle-detection teams.
- Supervised architecture, design and implementation during the redesign of mapping, 3D map formats and localization modules. Explained technical papers, coordinated technical direction across teams, and communicated complex software decisions. Technical debriefing following technical sessions.
- Investigated safety and compliance static code analysis tools like TrustInSoft, and standards Autosar and Misra.
- Technologies: C++, Eigen, FLANN, OpenNURBS, OpenMP, RTMaps, ZeroMQ, TCP/IP, CMake, Conan, JSON, SQL, UML, ParaView, MeshLab, TeamCity, Git, GitLab.

Philog – Software Engineering Consultant | Ile-de-France, France | may 2014 – december 2018

- **at Carestream Dental – Dental health | september 2015 – december 2018**
- Developed a 3D human-face reconstruction module covering photogrammetry, feature detection and matching, structure from motion, bundle adjustment, camera poses, point clouds, densification, regions of interest, meshes, mesh cleaning, loop removal, textures, graphs, clustering, PCA, camera data, and DICOM. Exploited methods and papers on surface reconstruction, image quality improvements, multiview stereo vision, and mathematical programming.
- Integrated multi-view stereo libraries and developed enhancements for desktop 3D software used in dental prosthesis design.
- Optimized processing and resource usage through parallelization, memory optimization, and sparse matrices.
- Maintained C and Python based command-line interfaces for critical legacy patient dbase database files.
- Refactored code, fixed defects, developed unit tests, and supported continuous integration.
- Technologies: C++, Boost, wxWidgets, VTK, MeshLab, OpenCV, Eigen, FLANN, Ceres, Dlib, OpenMP, CUDA, CMake, CodeBase, C, Python, Cppcheck, Visual Studio, Doxygen, Git, Targetprocess, TestTrack, Jenkins, scripts, Windows.
- **at Tech'Advantage – Geosciences | may 2014 – august 2015**
- Developed improvements to a subsurface-layer decompaction. Exploited methods and papers on finite elements, nonlinear elasticity, and nonlinear least-squares.
- Developed 3D visualization features for faults, regions, horizons, strains, and deformations.
- Optimized processing and resource usage through parallelization, memory optimization, and containers.
- Conducted code reviews, fixed defects, and developed unit and integration tests.
- Technologies: C++, Qt, STL, Boost, GoCAD, Code_Aster, Visual Studio, Makefile, XML, Jira, Collaborator, Cppcheck, Valgrind, SVN, Jenkins, Windows, Red Hat.

OptoPartner – Software Engineering Consultant | Ile-de-France, France | july 2012 – january 2014

- **at Leosphere – Lidar Wind Measurement**
- Converted complete software pipelines from LabVIEW and MATLAB to C++, involving lidar data, signal processing, instrumentation, trajectory planning, databases, networks, visualization, and command systems.
- Developed client/server software for operating lidar systems.
- Developed maximum-likelihood, least-squares, FFT, convolution, and region of interest. Ported processing code from MATLAB.
- Optimized and parallelized processing tasks. Ensured processing performance met the wind shear detection requirements.
- Developed visualizations including spectrograms, point clouds, and 2D graphics.
- Developed software for managing and controlling electronic and mechanical instruments.
- Managed databases, source code, testing, log analysis, installers, and embedded-system support.
- Wrote technical documentation, worked with instruments and prototypes, fixed defects, developed unit tests, and supported integration. Provided support during software acceptance testing.
- Integrated multiple technical components into production-oriented software pipelines.
- Technologies: C++, Qt, Boost, STL, GSL, FFTW, SQL, UML, XML, Eclipse, MySQL, SVN, Doxygen, Inno Setup, Debian packages, RS-232, USB, FTDI D2XX, Ethernet, TCP/IP, Telnet, Curl, MATLAB, Bugzilla, Windows, Ubuntu.

LITA - PostDoctoral Researcher – Financial Modeling | Metz, France | december 2011 – june 2012

- Worked on the Investment Fund Optimization project from the European Centre for Research in Financial Economics and Business Management, CEREFIGE.
- Conducted research on Markowitz-based portfolio selection, including Pareto sets, efficient frontiers, cardinality constraints, kurtosis, and skewness, using investment-fund data.
- Developed convexification techniques for solving regularized sparse modeling problems.
- Technologies: C++, CPLEX, MATLAB, MEX functions, Visual Studio.

INSA – Ministry-Funded PhD Researcher – Sparse Modeling | Rouen, France | oct. 2007 – nov. 2011

- Conducted doctoral research in DC programming, algorithms, sparse modeling, and data mining.
- Developed approaches and solutions for constrained and regularized sparse modeling problems.
- Published research and presented technical work at conferences and scientific events, including ICML.
- Conducted mathematics tutorials and assisted students with their projects.
- Technologies: C++, C, Python, MATLAB, R, Dlib, CPLEX, COIN-OR Xpress-MP, Weka.

CORIA - Engineering Intern – Computational Fluid Dynamics | Rouen, France | jun. 2007 – sept. 2007

- Extended a shock-tube gas-flow simulation from Navier–Stokes to BGK–Burnett. Performed compressible flow modeling, mathematical development and numerical integration. Applied WENO method and developed corresponding codes and their integration.
- Technologies: C, Fortran, Tecplot, Fedora, makefile.

INSA – Capstone Project – Cryptography | Rouen, France | 2007

- Developed balanced, highly nonlinear boolean functions using DC programming.
- Technologies: C, Xpress-MP.

UGB – 4-Year Bachelor Thesis – Statistics | St-Louis, Sénégal | 2005

- Developed robust estimators using influence-function and Newton-Raphson methods.
- Technologies: S, S-Plus.

Education

PhD, Applied Mathematics | INSA, Rouen, France | 2011
MSc, Mathematical and Software Engineering | INSA, Rouen, France | 2007
MSc, Fundamental and Applied Mathematics | INSA, Rouen, France | 2007
4-Year BSc, Applied Mathematics and Computer Science/Probability & Statistics | UGB, St-Louis, Sénégal | 2005